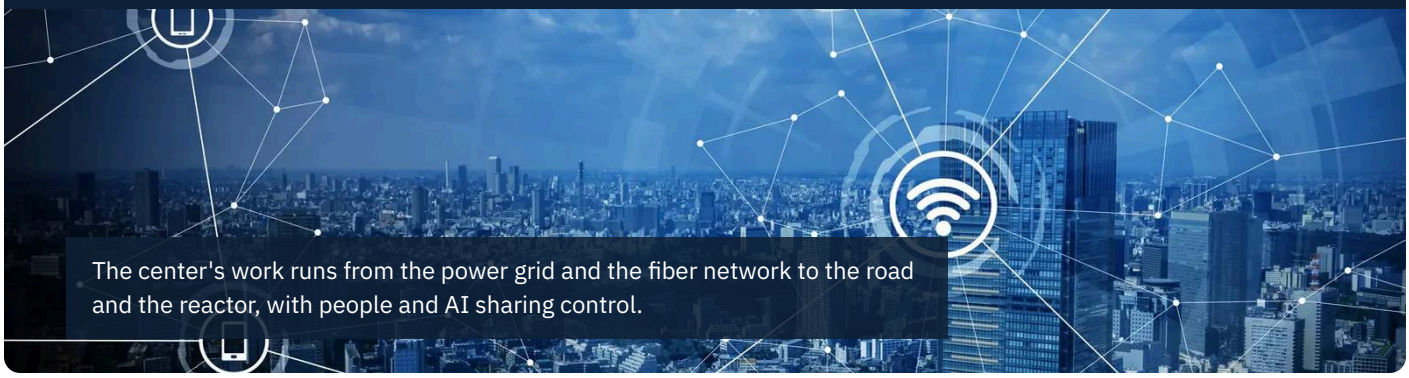
CENTER FOR SMART CYBER-PHYSICAL SYSTEMS · UMASS LOWELL VOLUME 1, NUMBER 2 | SEPTEMBER 2026

# SCyPS

The center's newsletter: research, people, and what comes next.



The center's work runs from the power grid and the fiber network to the road and the reactor, with people and AI sharing control.

FROM THE DIRECTOR

## Finished work, and the people behind it

Welcome to the second issue. September brought four new papers. Yan Luo's group, with Christopher Niezrecki and Murat Inalpolat, built a sensor network that listens to wind turbine blades to check their structural health. Arash Rezaee and Ryan McCann published FUSION, the open-source optical networking platform from our first issue's student story. Yuanchang Xie's group studied how drivers merge near work-zone signs. And Negin Yazdani Motlagh studied what computing employers ask for in job posts; her story is inside.

Our faculty spotlight is Lewis Tseng, who works on keeping groups of computers in agreement when some of them fail or are attacked. In August he won an NSF award to plan a shared data system for microscopy. Two of his four Ph.D. students are in the School of Education, co-advised with Mark Hsu, and that is the kind of collaboration we want more of.

On October 1, NSF's SUMMIT award began, and we are hiring a postdoctoral researcher to help build it. If you know someone who should apply, or a company that wants to test equipment against a real-time model of the Northeast grid, the last page says how to reach us.

*Vinod M. Vokkarane, Director*

### In this issue

1. From the director
2. The center: four thrusts, one loop
3. Faculty spotlight: Lewis Tseng
4. Student spotlight: Negin Yazdani Motlagh
5. September at the center, and what comes next

### By the numbers

**20** faculty, four colleges

**218** papers since 2019

**88** in journals

**16** doctoral students

THE CENTER

# Four thrusts, one loop

Cyber-physical systems sense the physical world, communicate what they sense, decide, and act. The center's research covers every link in that loop, and the four thrusts below are how its 20 faculty organize the work. Each thrust has two co-leads, a page on the center site, and a record of papers and awards that the site updates every week.

**Resilient energy and infrastructure**
Aghara, Xie, Lin, Vokkarane, Niezrecki, Inalpolat, Pagsuyoin, Chakrabarti, Srivastava, Zhao, Jain

**Next-generation networks and trusted devices**
Tseng, Arias, Vokkarane, Chigan, Akyurtlu, Ranasingha, Dempsey

**Cyber-physical AI**
Luo, Son, Vokkarane, Cao, Yu, Margala

**People in the loop**
Robinette, Evans, Hsu

## Active and new awards (20)

- SUMMIT
  National Science Foundation, Major Research Instrumentation Track 2, $2.0M
- ARPO-Sensor Fusion
  Massachusetts Technology Collaborative, Applied AI Models program, $625K
- BOND-AI
  SEMI/FlexTech, $500K
- ARPO: Autonomous Robotic Planning and Optimization
  U.S. Army, through Sherpa 6, $225K
- CyberCARED
  U.S. Department of Energy, $3.5M
- Massachusetts Advanced Nuclear and Fusion Energy Roadmaps
  Commonwealth of Massachusetts, Healey-Driscoll Administration
- Intercontinental Nuclear Institute (INI)
  International Atomic Energy Agency
- Advanced Technologies and Data Analytics for Safe, Smart, and Efficient Transportation (ASSET)
  Massachusetts Department of Transportation, $604K
- A Railroad Trespassing Detection and Dynamic Warning System Based on Artificial Intelligence and Edge Computing
  Massachusetts Department of Transportation, $750K
- Maine Advanced Signal Control and Connected Vehicle System for Safe, Efficient and Equitable Rural Transportation (MAST)
  U.S. Department of Transportation, ATCMTD, through MaineDOT, $150K
- Planning Federated AI-Ready Cyberinfrastructure for Advanced Microscopy and Imaging
  National Science Foundation, OAC Category III, $499K
- Towards Fault-tolerant Edge Computing for Cyber-Physical Systems
  National Science Foundation, CNS CAREER, $342K
- Improving Data Integrity for HPC Datasets using Sparsity Profile
  National Science Foundation, OAC Core, $600K
- Open-Source Research: Friendly Fedora and Podman
  Red Hat Inc., $400K+
- Resilient Smart Grids
  Navia Energy Inc., $60K+
- Phase II IUCRC at University of Massachusetts Lowell
  National Science Foundation, IUCRC-Indust-Univ Coop Res Ctr, $582K
- CAREER
  National Science Foundation, GOALI-Grnt Opp Acad Lia wIndus, EnvE-Environmental Engineering, $513K
- BioSPACE
  National Science Foundation, PFI-RP, $1.0M
- Unsupervised Deep PCCT Reconstruction for Human Extremity Imaging
  National Institutes of Health, NIBIB, $2.30M
- AI-based Cardiac CT
  National Institutes of Health, $2.4M

Full records at smartcyberphysical.org, where every paper and award links to its source.

FACULTY SPOTLIGHT

# Getting machines to agree, even when things go wrong

Lewis Tseng works on how groups of computers, from drone teams to blockchains, keep agreeing with each other when some of them fail or are attacked. In August, NSF funded his plan for a shared data system for microscopy that its users will help design.

Lewis Tseng, Associate Professor of Electrical and Computer Engineering, studies a question that comes up wherever one job is split across many computers: how do they agree on what happened, and keep agreeing when some of them crash, fall behind, or are attacked? He came to UMass Lowell in 2024 after positions at Clark University, Boston College, and Toyota InfoTechnology Center. He co-leads the center's next-generation networks and trusted devices thrust with Orlando Arias, and this year he received the ECE department's Teaching Excellence Award.

## The question

Drones searching an area together, edge computers running a control loop, and a blockchain recording transactions all depend on many machines staying consistent. If they stop agreeing, nothing built on top of them can be trusted. His NSF CAREER award, which began in 2024, looks at the basic coordination steps these systems rely on, and at how to keep them correct, and fast enough for real-time control, while someone is attacking the system.

## Recent work

In the past two years his group has published on multi-drone search coordinated through satellites and edge computers (IEEE GLOBECOM 2025), digital twins run as a service in hybrid clouds (IEEE ICHMS 2025), intrusion detection that keeps working when some detectors fail (ICSC 2025), and ways to replicate data reliably across machines (USENIX NSDI 2025; ACM SoCC 2025). In August he presented work at an ACM SIGCOMM workshop on timing control in quantum cloud systems, where entanglement decays if it isn't used in time.

## New in August: a data system its users help design

In August, NSF's Office of Advanced Cyberinfrastructure awarded Tseng $499,490 over two years to plan a federated, AI-ready data system for advanced microscopy and imaging. His co-PIs are Hsien-Yuan (Mark) Hsu of the School of Education and Yu-Tsun Shao of the University of Southern California.

The plan is for data to stay near the instruments and institutions that produce it, with shared services that help researchers find it, trace where it came from, trust it, and reuse it with AI tools. The team will design it with the scientists who would use it: first helping them picture what such a system could do, then studying how they work and what worries them, then designing the architecture, operating



Lewis Tseng, Associate Professor of Electrical and Computer Engineering.

### At a glance

- Co-lead, next-generation networks and trusted devices thrust
- NSF CAREER award, fault-tolerant edge computing (2024)
- PI, NSF planning award for federated AI-ready microscopy data (2026, $499,490)
- Co-PI, SUMMIT
- ECE Department Teaching Excellence Award (2026)
- Four Ph.D. students, two in the School of Education

model, and responsible AI practices together. Each lab keeps control of its own data, which makes it the same kind of trust problem Tseng works on in distributed systems.

## Engineering and education in one group

His group spans two colleges. Kritee Neupane, a computer engineering Ph.D. student, works on networked digital twins, blockchain provenance, and fault tolerance. Layann Shaban began her Ph.D. in 2026 after co-authoring his drone papers. Negin Yazdani Motlagh and Youlim Lee are Education Ph.D. students co-advised with Hsu: Negin studies AI and computational thinking in STEM education, and Youlim studies AI literacy and the ethical use of AI in research.

## Why it matters to the center

Grid controllers, autonomous robots, and the networks between them all depend on devices agreeing on what is true, sometimes while an attacker tries to split them. That is the problem Tseng works on, and it is why he co-leads the trusted devices thrust and is a Co-PI on SUMMIT, the center's three-university grid testbed.

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STUDENT SPOTLIGHT

# What employers ask for, in their own words

### Negin Yazdani Motlagh went through 82,624 job posts from 2012 to 2025 to see which soft skills computing employers ask for, and how often.



**Negin Yazdani Motlagh**
Ph.D. student, Research and Evaluation in Education, joined 2024; co-advised by Hsien-Yuan (Mark) Hsu

Advised by Hsien-Yuan (Mark) Hsu and Lewis Tseng, School of Education and Electrical and Computer Engineering

Engineering programs are asked to teach communication, teamwork, and problem solving alongside technical skills. Negin Yazdani Motlagh, a Ph.D. student in Research and Evaluation in Education since 2024, wanted to know which of those skills employers mention when they hire. She is co-advised by Mark Hsu of the School of Education and Lewis Tseng of Electrical and Computer Engineering, and her research focuses on artificial intelligence and computational thinking in STEM education.

## The data

She used the monthly "Who Is Hiring" threads on Hacker News, run by the startup accelerator Y Combinator, where companies post short job notices. Because the posts are short, what an employer includes says something about what it cares about. The study covers 82,624 computing and computer engineering postings from 2012 to 2025.

## The method

Each post was checked against a dictionary of soft skills sorted into three groups: interpersonal and leadership skills, personal effectiveness, and conceptual or thinking skills. Two measures follow each group over time: coverage, the share of posts that mention it, and intensity, the average number of mentions per post.

## What she found

Interpersonal and leadership skills came first on both measures, for every kind of job and in every year, with personal effectiveness second. Conceptual and thinking skills showed up less often, but they never dropped out, and they were more common in AI jobs. The paper offers these trends as a reference for instructors deciding how their computing and computer engineering courses handle professional skills.

## How the work grew

The study started as a work-in-progress paper at the 2025 IEEE Frontiers in Education Conference, covering 2012 to 2024, written with Kritee Neupane, a computer engineering Ph.D. student in Tseng's group. The full study, with Hsu and Tseng, was published in September in the Computers in Education Journal (CoED) of the American Society for Engineering Education. She also co-wrote a 2026 ACM Sustainability Week paper with Tseng on the energy cost of how blockchains reach agreement.

## Why it matters to the center

The center trains engineers through its doctoral students, its Red Hat-sponsored capstone, and its Open-Source Club, and Hsu leads its education and workforce work. Negin's study gives that work a measured starting point: what employers actually write when they hire.

## The papers

- Harmonizing Computing and Computer Engineering Education With Workforce Soft-Skill Signals: A Longitudinal Study of Short-Form Job Posts (2012–2025)
  Computers in Education Journal (CoED), vol. 15, no. 4, Sept. 2026; with H.-Y. Hsu and L. Tseng. doi:10.18260/b4d7-9a-69556
- WIP: What Employers Want: A Data-Driven Analysis of Soft Skill Trends in Computer Science Job Postings (2012-2024)
  IEEE Frontiers in Education Conference (FIE), Nov. 2025; with K. Neupane, L. Tseng, and H.-Y. Hsu. doi:10.1109/FIE63693.2025.11328524
- Green, Trust-Minimizing, and Sybil-Resistant? Rethinking How Blockchains Agree from an Energy Perspective
  ACM Sustainability Week, June 2026; with L. Tseng. doi:10.1145/3765611.3815150

THIS MONTH

# September 2026 at the center

## New papers

C. Pozzi, C. Ng, S. Lyon, Y. Luo, C. Niezrecki, M. Inalpolat. **A Novel Distributed Sensor Network to Assess Wind Turbine Blade Acoustics for Structural Health Monitoring**. *Wind Energy*, vol. 29, no. 10, Sept. 2026. doi:10.1002/we.70148

A. Rezaee, R. McCann, V. M. Vokkarane. **FUSION: A Unified Benchmarking Framework for Reproducible Optical Network Research**. *IEEE/Optica Journal of Optical Communications and Networking*, vol. 18, no. 9, pp. D90-D105, Sept. 2026 (Special Issue on Benchmarking in Optical Networks). doi:10.1364/JOCN.593123

N. Yazdani Motlagh, H. Y. Hsu, L. Tseng. **Harmonizing Computing and Computer Engineering Education With Workforce Soft-Skill Signals: A Longitudinal Study of Short-Form Job Posts (2012–2025)**. *CoED*, vol. 15, no. 4, Sept. 2026. doi:10.18260/b4d7-9a-69556

X. Yan, Y. Xie, Z. Bhuyan, B. Xiang, G. Wu, M. Shirazi. **Merging behavior under varying work zone sign scenarios: A heterogeneity-based analysis**. *Accident Analysis & Prevention*, vol. 235, art. 108614, Sept. 2026. doi:10.1016/j.aap.2026.108614

## Milestones

The FUSION benchmarking framework paper appears in JOCN's special issue on benchmarking in optical networks, followed in October by a QoT-aware grooming paper for multi-band SDM networks.

## Coming up

- **October 1** The NSF SUMMIT award began. The search for its postdoctoral researcher is open.
- **October 1** ARPO-Sensor Fusion, funded by the Massachusetts Technology Collaborative, began at UMLARC.
- **January 2027** BOND-AI begins: a SEMI/FlexTech project, led by Alkim Akyurtlu, on qualifying printed electronics that must survive 500 °C.
- **2026 to 2027** SUMMIT opens to collaborators from utilities, vendors, agencies, and other universities.
- **2028** IEEE HPSR 2028 at South Asian University, New Delhi, with the director as TPC Co-Chair.

## Work with us

Doctoral applicants: write to the faculty member whose work matches yours and copy the director. Companies and agencies: the laboratories page lists what each facility can offer, and the SUMMIT testbed opens to collaborators in 2026 to 2027.

## Support the center

Gifts fund student travel, seed projects, and the capstone program. Give at smartcyberphysical.org, or write to the director.

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